

EN 503: Energy Storage Technologies

Course Code	EN 503
Course Name	Energy Storage Technologies
L-T-P-C	3-0-0-3
Prerequisites	None
Intended for	UG/PG
Distribution	Elective
Mutual Exclusion	None
One-time approval	18th BoA (Sep 2017); 14th Senate (Oct 2017)
Revised	44th BoA (Nov 2021) - syllabus revised
Recoded to	MT 501 by the 57th BoA (Oct 2024)

Course Contents

- **Energy storage systems overview**
Scope of energy storage, needs and opportunities in energy storage, Technology overview and key disciplines, comparison of time scale of storages and applications, Energy storage in the power and transportation sectors. Importance of energy storage systems in electric vehicles, Current electric vehicle market. [5 Lectures]
- **Thermal storage system**
heat pumps, hot water storage tank, solar thermal collector, application of phase change materials for heat storage-organic and inorganic materials, efficiencies, and economic evaluation of thermal energy storage systems. [6 Lectures]
- **Chemical storage system**
Hydrogen, methane etc., concept of chemical storage of solar energy, application of chemical energy storage system, advantages and limitations of chemical energy storage, challenges, and future prospects of chemical storage systems. [5 Lectures]
- **Electromagnetic storage systems**
double layer capacitors with electrostatically charge storage, superconducting magnetic energy storage (SMES), concepts, advantages and limitations of electromagnetic energy storage systems, and future prospects of electrochemical storage systems. [5 Lectures]
- **Electrochemical storage system** [11 Lectures]
 - a **Batteries**: Working principle of battery, primary and secondary (flow) batteries, battery performance evaluation methods, major battery chemistries and their voltages- Li-ion battery & Metal hydride battery vs lead-acid battery.
 - b **Supercapacitors**: Working principle of supercapacitor, types of supercapacitors, cycling and performance characteristics, difference between battery and supercapacitors, Introduction to Hybrid electrochemical supercapacitors.
 - c **Fuel cell**: Operational principle of a fuel cell, types of fuel cells, hybrid fuel cell-battery systems, hybrid fuel cell-supercapacitor systems.
- **Module VI**
Battery design for transportation, Mechanical Design and Packaging of Battery Packs for Electric Vehicles, Advanced Battery-Assisted Quick Charger for Electric Vehicles, Charging Optimization Methods for Lithium-Ion Batteries, Thermal run-away for battery systems, Thermal management of battery systems, State of Charge and State of Health Estimation Over the Battery Lifespan, Recycling of Batteries from Electric Vehicles. [10 Lectures]

Text Books

1. Frank S. Barnes and Jonah G. Levine, **Large Energy Storage Systems Handbook** (Mechanical and Aerospace Engineering Series), CRC press, 2011.
2. Robert A. Huggins, **Energy storage**, Springer Science & Business Media, 2010.
3. Ralph Zito, **Energy storage: A new approach**, Wiley, 2010.

References

1. Pistoia, Gianfranco, and Boryann Liaw, **Behaviour of Lithium-Ion Batteries in Electric Vehicles: Battery Health, Performance, Safety, and Cost**, Springer International Publishing AG, 2018.
2. Robert A. Huggins, **Energy storage**, Springer Science & Business Media, 2010.